

Why the Bot Fails

A03 Reflection | GPT-4.1, the Worst Mathematician Ever

This bot can fail because a language model produces text by predicting likely continuations, rather than executing multiplication with an exact integer arithmetic engine. A prompt that says “return only an integer” can make the answer easier to compare, but it does not turn the model into a calculator. As the repeated-squaring loop continues, the numbers grow extremely quickly: after i squarings, the result is n raised to 2^i (for positive n). Long decimal strings are harder to reproduce reliably, and a single wrong digit makes the exact comparison fail.

The experiment also measures more than arithmetic skill. The API may return an explanation, formatting, or an error instead of a plain integer. The notebook treats those responses as unusable and records their details separately. A request can fail because of a network or API problem, so that is not evidence that the model made a mathematical mistake. The log distinguishes a wrong answer from a response that could not be parsed.

Python calculates the reference answer using arbitrary-precision integers, so it does not silently overflow at ordinary assignment sizes. The program compares the model's parsed integer with that exact value after every step. The humorous line is triggered by a mismatch, and its tone escalates after repeated errors. This reaction is part of the bot's design; it does not repair the model's reasoning.

Results should be interpreted cautiously. Model outputs can vary between calls, even when temperature is set to zero, and one run cannot establish a general failure rate. The three challenge combinations in the notebook are candidate tests, not guaranteed failures. This workspace cannot access the private Colab secret shown in the assignment screenshot, so the included starter log does not pretend to contain observed API outputs. Run the notebook with an authorized key, retain three combinations that actually fail, and export the generated log. If every case succeeds, increase the iteration count and report that outcome honestly. A fair conclusion is that exact arithmetic is unreliable enough to verify independently, not that every answer will be wrong.